R-107, First Floor, Metro Tower, Vijay Nagar, Indore (M.P.) 452010

AI Agent Development — Practical Course Curriculum

Cypher Analytica — Build AI systems that use tools and take controlled action

This course teaches learners to develop AI agents that interpret a goal, select appropriate tools, use their results, and determine the next step within defined limits.

Core concept: In an AI-assisted workflow, predefined rules largely control the sequence. An AI agent can choose some actions dynamically—for example, deciding whether to search a knowledge base, query a database, or ask the user for missing information.

Duration: 50 hours of guided training + 25 hours of assignments and project work
Audience: Students, Python developers, data professionals, and learners who have completed AI-assisted workflow automation
Prerequisites: Basic Python, functions, dictionaries, JSON, APIs, and introductory LLM concepts.

By the end, learners should be able to build, evaluate, and deploy a tool-using agent with controlled access, reliable state management, and human approval for consequential actions.

Module-wise curriculum

Module	Hours	Topics covered	Practical activity
1. Foundations of AI Agents	3	Agents versus chatbots and workflows; goals; instructions; tools; state; environment; autonomy limits; when an agent is unnecessary	Define an agent's responsibilities, tools, boundaries, and success criteria
2. Python and API Foundations	4	Functions; type hints; JSON; HTTP requests; environment variables; exception handling; API authentication; reusable tool functions	Create functions for searching sample course records and checking batch availability
3. LLM Integration and Structured Responses	4	Messages and context; model connection; structured outputs; schemas; validation; token usage; handling missing information	Convert a user request into a validated task specification

Module	Hours	Topics covered	Practical activity
4. Tool Calling and Execution	6	Tool definitions; argument schemas; tool selection; argument validation; execution; returning results; read versus write tools; handling unavailable tools	Build an assistant that chooses between a calculator, course search, and timetable lookup
5. Building the Agent Execution Loop	5	Request → model decision → tool execution → observation → next decision; completion conditions; step limits; timeouts; repeated-action detection; clarification	Build a bounded agent that completes a task using multiple tool calls
6. State, Memory, and Context Management	4	Conversation history; working state; persistent user preferences; session isolation; summarisation; context limits; retention and deletion	Build a multi-turn counsellor that remembers preferences within the correct session
7. Retrieval and Knowledge-Grounded Agents	5	Document preparation; chunking; embeddings concepts; retrieval; metadata filtering; source references; freshness; handling conflicting or absent information	Build an agent that answers course questions using an approved knowledge base
8. Planning and Human Approval	4	Task decomposition; adapting to tool results; clarification; approval gates; previews of proposed actions; cancellation; recovery	Let an agent propose a counselling appointment and request approval before booking
9. Security, Reliability, and Evaluation	6	Prompt injection; untrusted tool output; least privilege; credential protection; duplicate prevention; retries; test sets; task success; tool-call correctness; latency and cost	Test malicious documents, invalid tool arguments, unavailable services, and repeated requests

Module	Hours	Topics covered	Practical activity
10. Deployment and Capstone Project	9	Application interface; configuration; logging; execution traces; monitoring; deployment; documentation; end-to-end evaluation	Deploy and demonstrate a controlled admission counselling agent
Total	**50**		

Practical assignments

1. **Tool-using assistant:** Select a calculator or record-lookup tool based on the request.
2. **Course information agent:** Answer questions from approved documents with source references.
3. **Database enquiry agent:** Retrieve information through constrained, read-only functions.
4. **Multi-turn counselling agent:** Collect preferences and ask for missing details without mixing user sessions.
5. **Approval-based action agent:** Prepare an appointment or email and execute only after approval.
6. **Agent evaluation exercise:** Measure task completion, factual accuracy, tool selection, and failure handling.

Capstone: Cypher Analytica Admission Counselling Agent

Scenario: A prospective student asks about learning options. The agent clarifies their goals, retrieves approved information, explains suitable options, and offers to arrange a counselling session.

Capability	Expected behaviour
Understand the learner	Ask about goals, experience, availability, and preferred learning mode
Search course information	Retrieve relevant details from the approved knowledge base

Capability	Expected behaviour
Explain course suitability	Connect the learner's stated needs to documented course content
Check availability	Use a timetable tool to retrieve available counselling slots
Prepare an action	Show the proposed booking or follow-up message for review
Request approval	Obtain explicit confirmation before booking or sending
Record the outcome	Save only the necessary details through a validated tool
Escalate uncertainty	Refer missing or conflicting information to a human counsellor

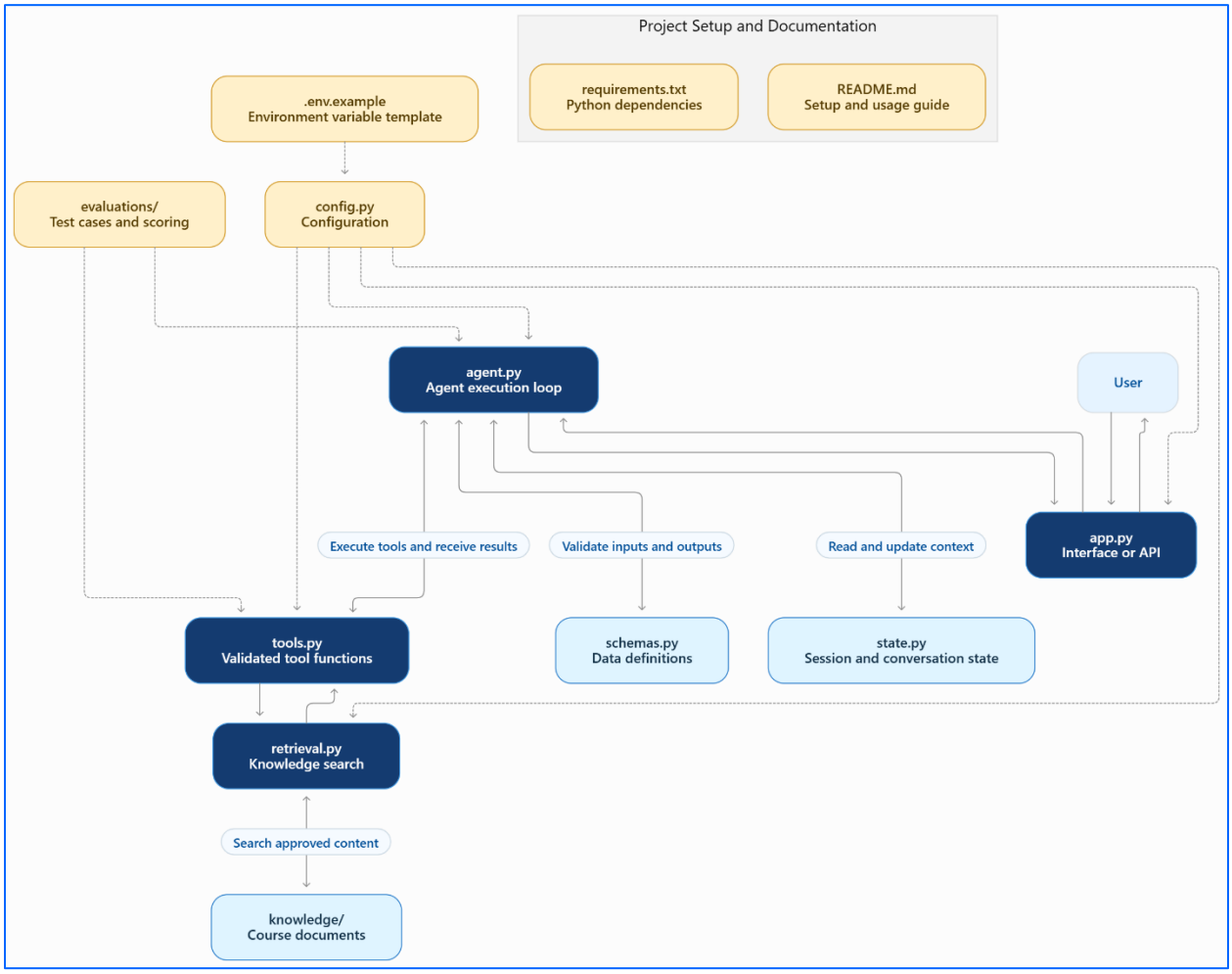
The agent must not invent fees, schedules, certifications, or placement guarantees. When tools fail, it should explain the limitation and offer an appropriate next step.

Required deliverables:

- Agent specification defining goals, tools, permissions, and stopping rules.
- Working application with documented tool schemas.
- Approved sample knowledge base and synthetic enquiry records.
- Evaluation dataset covering routine, ambiguous, and adversarial requests.
- Test report, execution traces, setup guide, and demonstration.

R-107, First Floor, Metro Tower, Vijay Nagar, Indore (M.P.) 452010

Suggested project structure



Assessment and certification

Assessment	Weight
Practical assignments	25%
Reliability and security evaluation	25%
Capstone project	40%
Demonstration and viva	10%